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AIUB, Computer Graphics Notes

COMPUTER GRAPHICS

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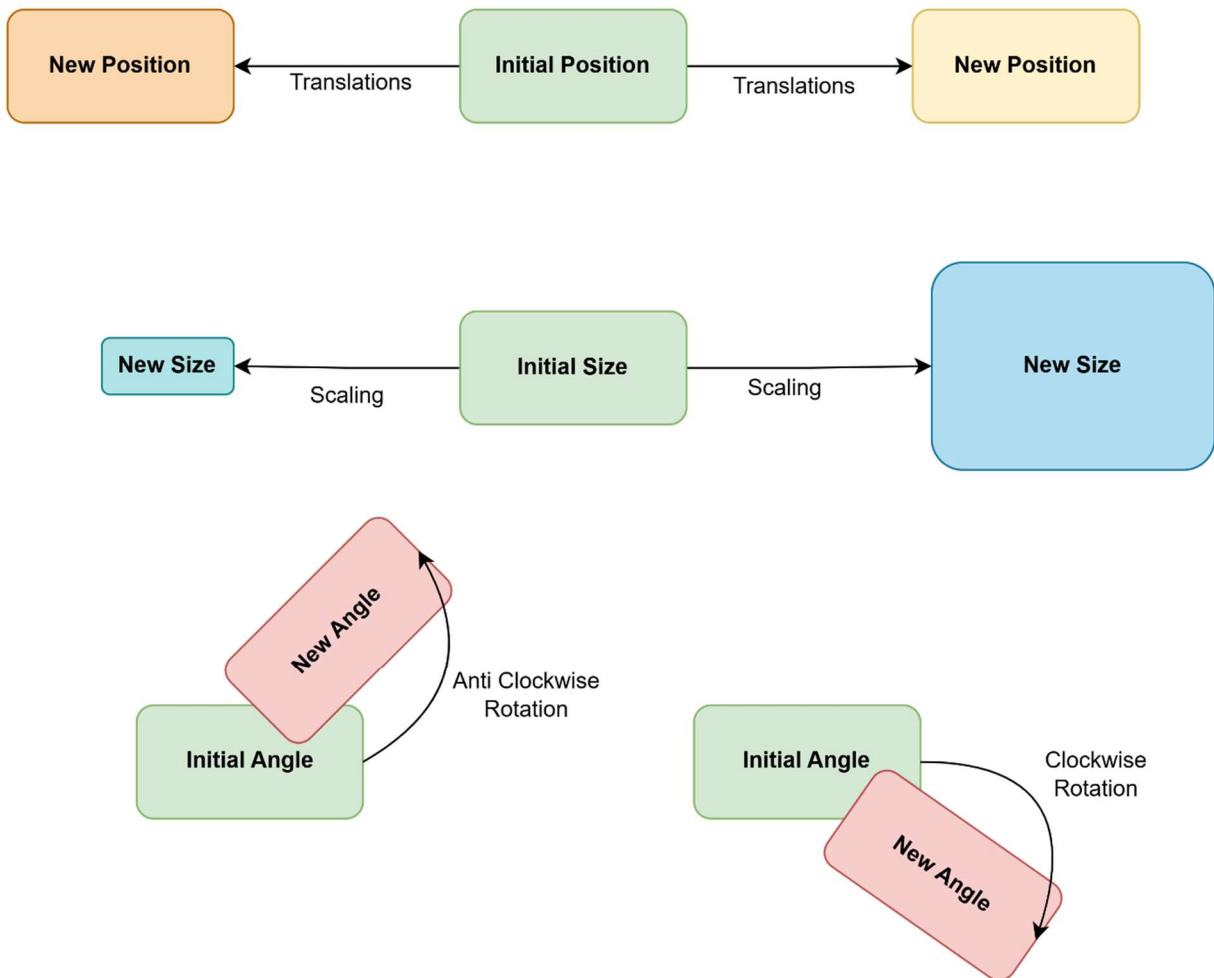
2D-TRANSFORMATION

Transformation

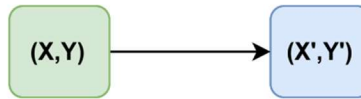
➔ A **transformation** is a mathematical operation that changes the position, size, orientation, or shape of an object or coordinate system.

In computer graphics, transformations are used to manipulate objects or images, for example by moving (translating), rotating, scaling (resizing), or reflecting them.

Types of 2D-Transformation



Translations



Formula

Equation Form

$$x' = x + tx$$

$$y' = y + ty$$

Where,

$tx = ty = \text{translation factor}$

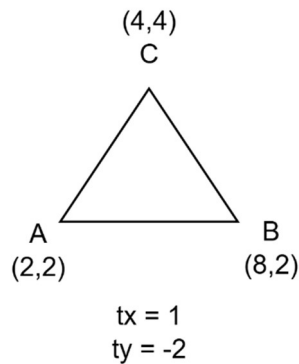
Matrix Form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x_1 + tx & x_2 + tx & x_3 + tx & \dots & x_n + tx \\ y_1 + ty & y_2 + ty & y_2 + ty & \dots & y_n + ty \end{bmatrix}$$

Where,

$tx = ty = \text{translation factor}$

✂ **Problem 1:** Given object with coordinate ABC & translation factor tx, ty. Find the coordinate of the object after applying translations using equations & Matrix Form



➔ Is given,

$$tx = 1$$

$$ty = -2$$

We Know,

Using equations:

A	B	C
$x' = x + tx = 2 + 1 = 3$ $y' = y + ty = 2 + (-2) = 0$	$x' = x + tx = 8 + 1 = 9$ $y' = y + ty = 2 + (-2) = 0$	$x' = x + tx = 4 + 1 = 5$ $y' = y + ty = 4 + (-2) = 2$
A = (3, 0)	B = (9, 0)	C = (5, 2)

Using Matrix:

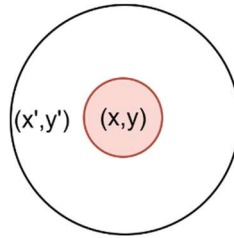
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x_1 + tx & x_2 + tx & x_3 + tx \dots x_n + tx \\ y_1 + ty & y_2 + ty & y_2 + ty \dots y_n + ty \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 + 1 & 8 + 1 & 4 + 1 \\ 2 + (-2) & 2 + (-2) & 4 + (-2) \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 9 & 5 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\therefore \mathbf{A = (3, 0), B = (9, 0), C = (5, 2)}$$

Scaling



Formula

Equation Form

$$x' = x \times Sx$$

$$y' = y \times Sy$$

Where,

$Sx = Sy = \text{Scaling factor}$

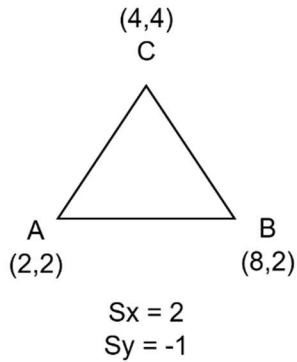
Matrix Form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

Where,

$Sx = Sy = \text{Scaling factor}$

✂ **Problem 2:** Given object with coordinate ABC & Scaling factor S_x , S_y . Find the coordinate of the object after applying Scaling using equations & Matrix Form



→ Is given,

$$S_x = 2$$

$$S_y = -1$$

We Know,

Using equations:

A	B	C
$x' = x \times S_x = 2 \times 2 = 4$ $y' = y \times S_y = 2 \times (-1) = -2$	$x' = x \times S_x = 8 \times 2 = 16$ $y' = y \times S_y = 2 \times (-1) = -2$	$x' = x \times S_x = 4 \times 2 = 8$ $y' = y \times S_y = 4 \times (-1) = -4$
$A = (4, -2)$	$B = (16, -2)$	$C = (8, -4)$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 2 & 8 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

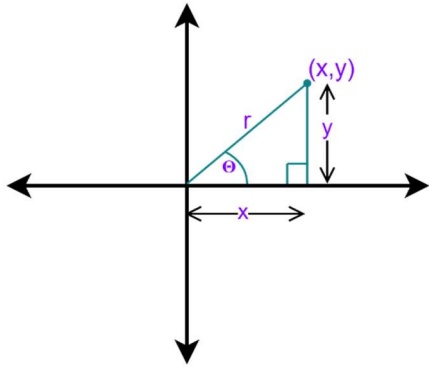
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & 16 & 8 \\ -2 & -2 & -4 \end{bmatrix}$$

$$\therefore A = (4, -2), B = (16, -2), C = (8, -4)$$

Rotations

Anti-Clockwise

Derivations

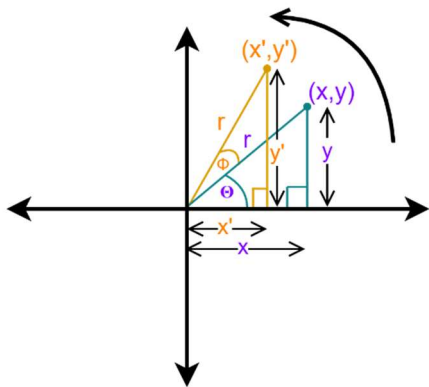


$$\cos \theta = \frac{x}{r}$$

$$\therefore x = r \times \cos \theta$$

$$\sin \theta = \frac{y}{r}$$

$$\therefore y = r \times \sin \theta$$



$$\cos(\theta + \phi) = \frac{x'}{r}$$

$$\therefore x' = r \times \cos(\theta + \phi)$$

$$x' = r \times [\cos(\theta) \cdot \cos(\phi) - \sin(\theta) \cdot \sin(\phi)]$$

$$x' = r \cdot \cos(\theta) \cdot \cos(\phi) - r \cdot \sin(\theta) \cdot \sin(\phi)$$

$$x' = x \cdot \cos(\phi) - y \cdot \sin(\phi)$$

$$\sin(\theta + \phi) = \frac{y'}{r}$$

$$\therefore y' = r \times \sin(\theta + \phi)$$

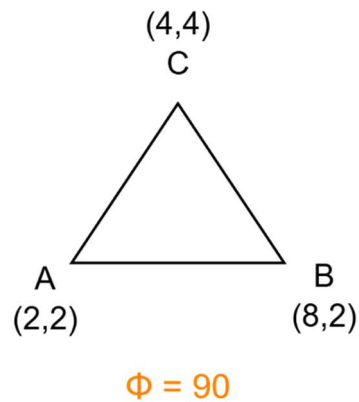
$$y' = r \times [\sin(\theta) \cdot \cos(\phi) + \sin(\phi) \cdot \cos(\theta)]$$

$$y' = r \cdot \sin(\theta) \cdot \cos(\phi) + r \cdot \sin(\phi) \cdot \cos(\theta)$$

$$y' = y \cdot \cos(\phi) + x \cdot \sin(\phi)$$

Formula
Equation Form
$x' = x.Cos(\Phi) - y.Sin(\Phi)$ $y' = x.Sin(\Phi) + y.Cos(\Phi)$ Where, $\Phi = \text{Rotation factor}$
Matrix Form
$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} Cos(\Phi) & -Sin(\Phi) \\ Sin(\Phi) & Cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3... & x_n \\ y_1 & y_2 & y_3... & y_n \end{bmatrix}$ Where, $\Phi = \text{Rotation factor}$

🔗 **Problem 3:** Given object with coordinate ABC & Rotation factor $\Phi = 90^\circ$. Find the coordinate of the object after applying **anti clockwise** Rotating using equations & Matrix Form



→ Is given,

$$\Phi = 90^\circ$$

We Know,

Using equations:

A	B	C
$x' = x \cdot \cos(\Phi) - y \cdot \sin(\Phi)$ $x' = 2 \cdot \cos(90) - 2 \cdot \sin(90)$ $x' = 2 \times 0 - 2 \times 1 = -2$ $y' = x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = 2 \cdot \sin(90) + 2 \cdot \cos(90)$ $y' = 2 \times 1 + 2 \times 0 = 2$	$x' = x \cdot \cos(\Phi) - y \cdot \sin(\Phi)$ $x' = 8 \cdot \cos(90) - 2 \cdot \sin(90)$ $x' = 8 \times 0 - 2 \times 1 = -2$ $y' = x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = 8 \cdot \sin(90) + 2 \cdot \cos(90)$ $y' = 8 \times 1 + 2 \times 0 = 8$	$x' = x \cdot \cos(\Phi) - y \cdot \sin(\Phi)$ $x' = 4 \cdot \cos(90) - 4 \cdot \sin(90)$ $x' = 4 \times 0 - 4 \times 1 = -4$ $y' = x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = 4 \cdot \sin(90) + 4 \cdot \cos(90)$ $y' = 4 \times 1 + 4 \times 0 = 4$
$A = (-2, 2)$	$B = (-2, 8)$	$C = (-4, 4)$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\Phi) & -\sin(\Phi) \\ \sin(\Phi) & \cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

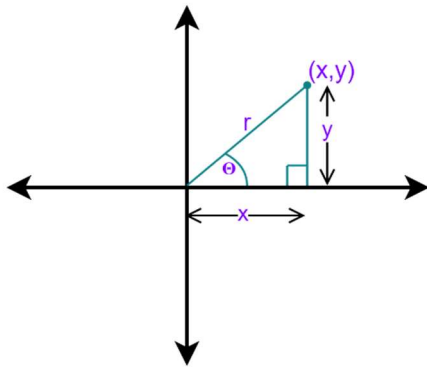
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(90) & -\sin(90) \\ \sin(90) & \cos(90) \end{bmatrix} \times \begin{bmatrix} 2 & 8 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 8 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -2 & -2 & -4 \\ 2 & 8 & 4 \end{bmatrix}$$

$$\therefore A = (-2, 2), B = (-2, 8), C = (-4, 4)$$

Derivations

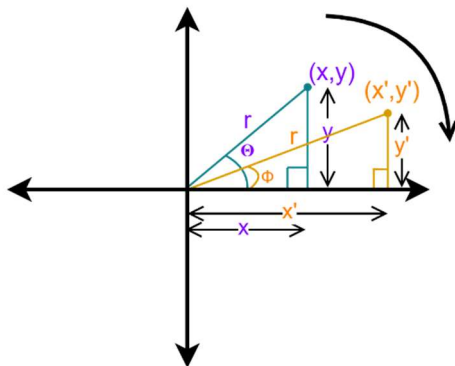


$$\cos \theta = \frac{x}{r}$$

$$\therefore x = r \times \cos \theta$$

$$\sin \theta = \frac{y}{r}$$

$$\therefore y = r \times \sin \theta$$



$$\cos(\theta - \phi) = \frac{x'}{r}$$

$$\therefore x' = r \times \cos(\theta - \phi)$$

$$x' = r \times [\cos(\theta) \cdot \cos(\phi) + \sin(\theta) \cdot \sin(\phi)]$$

$$x' = r \cdot \cos(\theta) \cdot \cos(\phi) + r \cdot \sin(\theta) \cdot \sin(\phi)$$

$$x' = x \cdot \cos(\phi) + y \cdot \sin(\phi)$$

$$\sin(\theta - \phi) = \frac{y'}{r}$$

$$\therefore y' = r \times \sin(\theta - \phi)$$

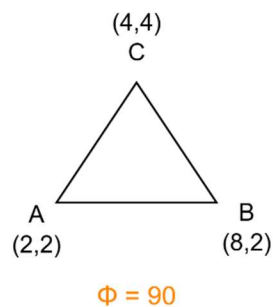
$$y' = r \times [\sin(\theta) \cdot \cos(\phi) - \sin(\phi) \cdot \cos(\theta)]$$

$$y' = r \cdot \sin(\theta) \cdot \cos(\phi) - r \cdot \sin(\phi) \cdot \cos(\theta)$$

$$y' = y \cdot \cos(\phi) - x \cdot \sin(\phi)$$

Formula
Equation Form
$x' = x.Cos(\Phi) + y.Sin(\Phi)$ $y' = -x.Sin(\Phi) + y.Cos(\Phi)$ Where, $\Phi = \text{Rotation factor}$
Matrix Form
$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} Cos(\Phi) & Sin(\Phi) \\ -Sin(\Phi) & Cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3... & x_n \\ y_1 & y_2 & y_3... & y_n \end{bmatrix}$ Where, $\Phi = \text{Rotation factor}$

✂ **Problem 4:** Given object with coordinate ABC & Rotation factor $\Phi = 90^\circ$. Find the coordinate of the object after applying **clockwise** Rotating using equations & Matrix Form



→ Is given,
 $\Phi = 90^\circ$

We Know,

Using equations:

A	B	C
$x' = x \cdot \cos(\Phi) + y \cdot \sin(\Phi)$ $x' = 2 \cdot \cos(90) + 2 \cdot \sin(90)$ $x' = 2 \times 0 + 2 \times 1 = +2$ $y' = -x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = -2 \cdot \sin(90) + 2 \cdot \cos(90)$ $y' = -2 \times 1 + 2 \times 0 = -2$	$x' = x \cdot \cos(\Phi) + y \cdot \sin(\Phi)$ $x' = 8 \cdot \cos(90) + 2 \cdot \sin(90)$ $x' = 8 \times 0 + 2 \times 1 = 2$ $y' = -x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = -8 \cdot \sin(90) + 2 \cdot \cos(90)$ $y' = -8 \times 1 + 2 \times 0 = -8$	$x' = x \cdot \cos(\Phi) + y \cdot \sin(\Phi)$ $x' = 4 \cdot \cos(90) + 4 \cdot \sin(90)$ $x' = 4 \times 0 + 4 \times 1 = +4$ $y' = -x \cdot \sin(\Phi) + y \cdot \cos(\Phi)$ $y' = -4 \cdot \sin(90) + 4 \cdot \cos(90)$ $y' = -4 \times 1 + 4 \times 0 = -4$
$A = (2, -2)$	$B = (2, -8)$	$C = (4, -4)$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\Phi) & \sin(\Phi) \\ -\sin(\Phi) & \cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

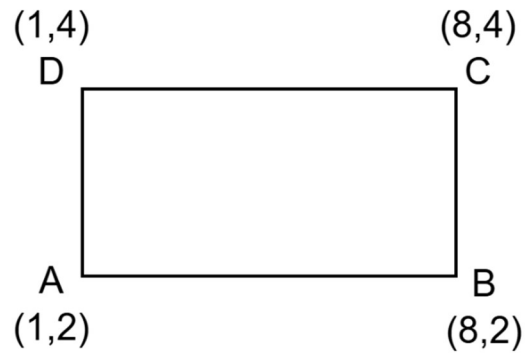
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(90) & \sin(90) \\ -\sin(90) & \cos(90) \end{bmatrix} \times \begin{bmatrix} 2 & 8 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 8 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 \\ -2 & -8 & -4 \end{bmatrix}$$

$$\therefore A = (2, -2), B = (2, -8), C = (4, -4)$$

✂ **Problem 5:** An object with coordinates ABCD is first translated with $tx = 1$ & $ty = 2$. The object is then again translated with $tx = 3$ & $ty = 3$. After that, the object is rotated 90° anti-Clockwise. After this the object scaled with $Sx = 2$ & $Sy = 2$. Finally again the object is rotated 90° Clockwise. Find the new coordinate of the object after transformations.



→ Is given,

$$tx = 1$$

$$ty = 2$$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x_1 + tx & x_2 + tx & x_3 + tx & \dots & x_n + tx \\ y_1 + ty & y_2 + ty & y_2 + ty & \dots & y_n + ty \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 + 1 & 8 + 1 & 8 + 1 & 1 + 1 \\ 2 + 2 & 2 + 2 & 4 + 2 & 4 + 2 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 9 & 9 & 2 \\ 4 & 4 & 6 & 6 \end{bmatrix}$$

$$\therefore A = (2, 4), B = (9, 4), C = (9, 6), D = (2, 6)$$

➔ Now,

$$tx = 3$$

$$ty = 3$$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x_1 + tx & x_2 + tx & x_3 + tx \dots x_n + tx \\ y_1 + ty & y_2 + ty & y_2 + ty \dots y_n + ty \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 + 3 & 9 + 3 & 9 + 3 & 2 + 3 \\ 4 + 3 & 4 + 3 & 6 + 3 & 6 + 3 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 5 & 12 & 12 & 5 \\ 7 & 7 & 9 & 9 \end{bmatrix}$$

$$\therefore A = (5, 7), B = (12, 7), C = (12, 9), D = (5, 9)$$

➔ Is given,

$$\Phi = 90^\circ \quad [\text{Anti-Clockwise}]$$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\Phi) & -\sin(\Phi) \\ \sin(\Phi) & \cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots x_n \\ y_1 & y_2 & y_3 \dots y_n \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(90) & -\sin(90) \\ \sin(90) & \cos(90) \end{bmatrix} \times \begin{bmatrix} 5 & 12 & 12 & 5 \\ 7 & 7 & 9 & 9 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 5 & 12 & 12 & 5 \\ 7 & 7 & 9 & 9 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -7 & -7 & -9 & -9 \\ 5 & 12 & 12 & 5 \end{bmatrix}$$

$$\therefore A = (-7, 5), B = (-7, 12), C = (-9, 12), D = (-9, 5)$$

→ Is given,

$$Sx = 2$$

$$Sy = 2$$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} -7 & -7 & -9 & -9 \\ 5 & 12 & 12 & 5 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -14 & -14 & -18 & -18 \\ 10 & 24 & 24 & 10 \end{bmatrix}$$

$$\therefore A = (-14, 10), B = (-14, 24), C = (-18, 24), D = (-18, 10)$$

→ Is given,

$$\Phi = 90^\circ \quad [\text{Clockwise}]$$

Using Matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\Phi) & \sin(\Phi) \\ -\sin(\Phi) & \cos(\Phi) \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \dots & x_n \\ y_1 & y_2 & y_3 \dots & y_n \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(90) & \sin(90) \\ -\sin(90) & \cos(90) \end{bmatrix} \times \begin{bmatrix} -14 & -14 & -18 & -18 \\ 10 & 24 & 24 & 10 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} -14 & -14 & -18 & -18 \\ 10 & 24 & 24 & 10 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 10 & 24 & 24 & 10 \\ 14 & 14 & 18 & 18 \end{bmatrix}$$

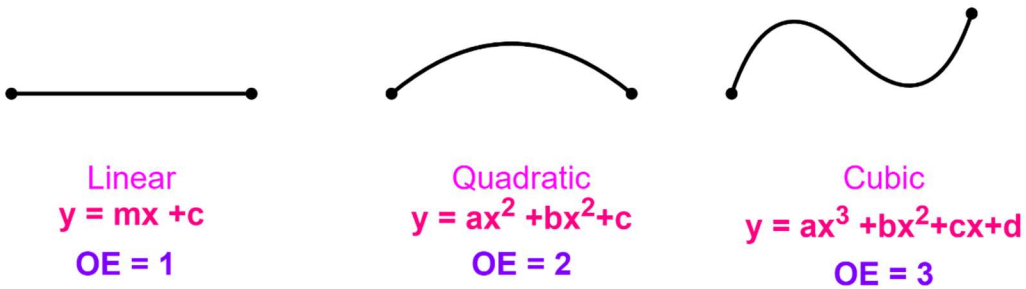
$$\therefore A = (10, 14), B = (25, 14), C = (24, 18), D = (10, 18)$$

∴ New coordinates:

$$\therefore A = (10, 14), B = (25, 14), C = (24, 18), D = (10, 18)$$

BEZIER CURVES

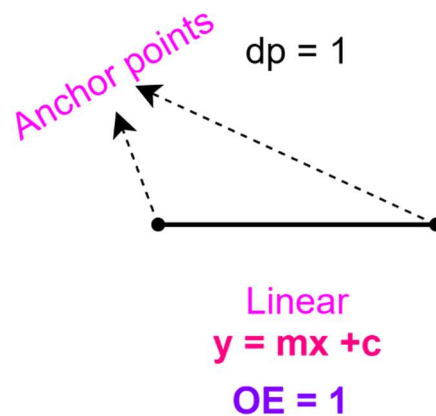
Bezier Curves



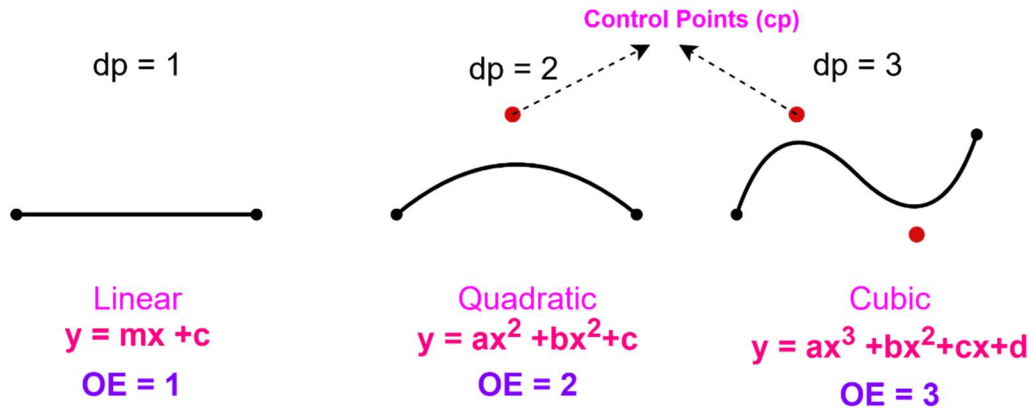
Degree of Polynomial (dp) = OE
OE = Order of Equations

Properties of Bezier Curves

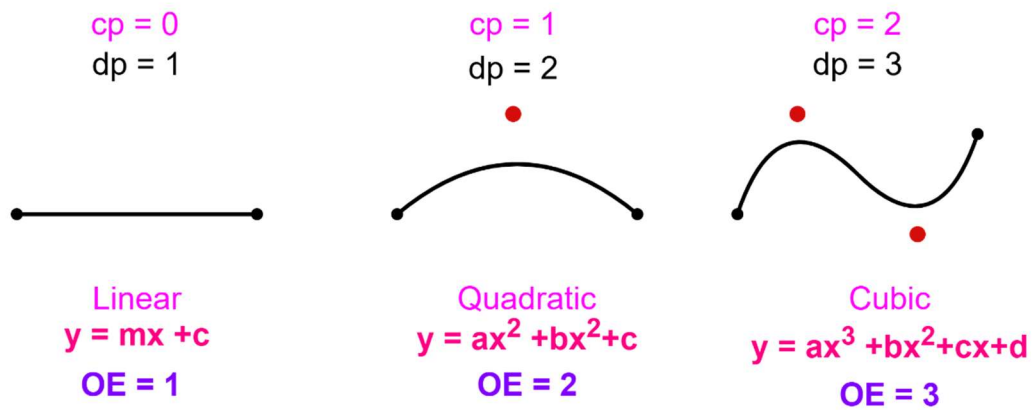
1. A Bezier Curves will always touch the first and last points which are known as **anchor points**.



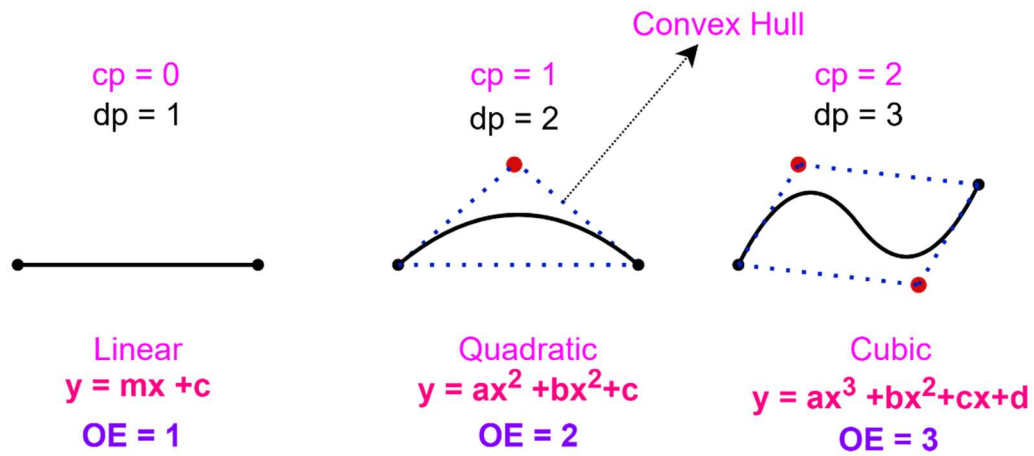
2. A Bézier curves are defined by a set of **special points known as control points**.



3. Degree of polynomial is always **one more than the total number of control points**.



4. A Bézier curves always **contain with in the convex hull** that is defined by the controls points & the anchor points.



5. A Bézier curves exhibit global Control Points

Blending Functions

$$P(u) = \sum_{i=0}^n P_i B_{i,n}(u)$$

Where,

$n = \text{degree of polynomial}$

$u = 0 \leq u \leq 1$

$B_{i,n}(u) = C(n, i) \cdot (u)^i \cdot (1 - u)^{n-i}$

$C(n, i) = \frac{n!}{i!(n-i)!}$

$$\therefore P(u) = \sum_{i=0}^n P_i \times \frac{n!}{i!(n-i)!} \times (u)^i \times (1 - u)^{n-i}$$

Problem 1: Given four points:

- $P_0 = (1,1)$
- $P_1 = (2,3)$
- $P_2 = (4,3)$
- $P_3 = (6,4)$

Use the Bernstein polynomial form to compute the point $P(u)$ on the **cubic Bézier curve**.

→ Is given,

Total points = 4

so, degree of polynomial $n = 3$

→ We know,

$$P(u) = \sum_{i=0}^3 P_i B_{i,n}(u)$$
$$= P_0 B_{0,3} + P_1 B_{1,3} + P_2 B_{2,3} + P_3 B_{3,3} \dots\dots\dots(i)$$

$$P_i B_{i,n} = P_i \times \frac{n!}{i! (n-i)!} \times (u)^i \times (1-u)^{n-i}$$

$$\therefore P_0 B_{0,3} = P_0 \times \frac{3!}{0! (3-0)!} \times (u)^0 \times (1-u)^{3-0} = P_0 \times (1-u)^3$$

$$\therefore P_1 B_{1,3} = P_1 \times \frac{3!}{1! (3-1)!} \times (u)^1 \times (1-u)^{3-1} = P_1 \times 3u \times (1-u)^2$$

$$\therefore P_2 B_{2,3} = P_2 \times \frac{3!}{2! (3-2)!} \times (u)^2 \times (1-u)^{3-2} = P_2 \times 3u^2 \times (1-u)^1$$

$$\therefore P_3 B_{3,3} = P_3 \times \frac{3!}{3! (3-3)!} \times (u)^3 \times (1-u)^{3-3} = P_3 \times u^3$$

Put those values in equation 1:

$$\therefore P(u) = P_0(1-u)^3 + P_1 3u(1-u)^2 + P_2 3u^2(1-u)^1 + P_3 u^3$$

$$\therefore X(u) = X_0(1-u)^3 + X_1 3u(1-u)^2 + X_2 3u^2(1-u)^1 + X_3 u^3 \dots\dots(ii)$$

$$\therefore Y(u) = Y_0(1-u)^3 + Y_1 3u(1-u)^2 + Y_2 3u^2(1-u)^1 + Y_3 u^3 \dots\dots(iii)$$

Put those values in X_0, X_1, X_2, X_3 Values in equation 2:

$$\therefore X(u) = 1(1-u)^3 + 2 \times 3u(1-u)^2 + 4 \times 3u^2(1-u)^1 + 6u^3$$

$$\therefore X(u) = (1-u)^3 + 6u(1-u)^2 + 12u^2(1-u)^1 + 6u^3 \dots\dots (iv)$$

Put those values in Y_0, Y_1, Y_2, Y_3 Values in equation 3:

$$\therefore Y(u) = 1(1-u)^3 + 3 \times 3u(1-u)^2 + 3 \times 3u^2(1-u)^1 + 4u^3$$

$$\therefore Y(u) = (1-u)^3 + 9u(1-u)^2 + 9u^2(1-u)^1 + 4u^3 \dots\dots (v)$$

So the point P(u) on the cubic Bézier curve:

u	X	Y
0.0	1	1
0.2	1.712	1.984
0.4	2.616	2.632
0.6	3.664	3.088
0.8	4.808	3.496
1.0	6	4

Problem 2: Given five points:

- $P_0 = (1,1)$
- $P_1 = (2,3)$
- $P_2 = (4,3)$
- $P_3 = (6,4)$
- $P_4 = (8,6)$

Use the Bernstein polynomial form to compute the point $P(u)$ on the **Bézier curve**.

→ Is given,

points = 5

so, degree of polynomial $n = 4$

→ We know,

$$P(u) = \sum_{i=0}^4 P_i B_{i,n}(u)$$

$$= P_0 B_{0,4} + P_1 B_{1,4} + P_2 B_{2,4} + P_3 B_{3,4} + P_4 B_{4,4} \dots \dots \dots (i)$$

$$P_i B_{i,n} = P_i \times \frac{n!}{i! (n-i)!} \times (u)^i \times (1-u)^{n-i}$$

$$\therefore P_0 B_{0,4} = P_0 \times \frac{4!}{0! (4-0)!} \times (u)^0 \times (1-u)^{4-0} = P_0 \times (1-u)^4$$

$$\therefore P_1 B_{1,4} = P_1 \times \frac{4!}{1! (4-1)!} \times (u)^1 \times (1-u)^{4-1} = P_1 \times 4u \times (1-u)^3$$

$$\therefore P_2 B_{2,4} = P_2 \times \frac{4!}{2! (4-2)!} \times (u)^2 \times (1-u)^{4-2} = P_2 \times 6u^2 \times (1-u)^2$$

$$\therefore P_3 B_{3,4} = P_3 \times \frac{4!}{3! (4-3)!} \times (u)^3 \times (1-u)^{4-3} = P_3 \times 4u^3 \times (1-u)^1$$

$$\therefore P_4 B_{4,4} = P_4 \times \frac{4!}{4! (4-4)!} \times (u)^4 \times (1-u)^{4-4} = P_4 \times u^4$$

Put those values in equation 1:

$$\therefore P(u) = P_0(1-u)^4 + P_1 4u(1-u)^3 + P_2 6u^2(1-u)^2 + P_3 4u^3(1-u)^1 + P_4 u^4$$

$$\therefore X(u) = X_0(1-u)^4 + X_1 4u(1-u)^3 + X_2 6u^2(1-u)^2 + X_3 4u^3(1-u)^1 + X_4 u^4 \dots \dots (ii)$$

$$\therefore Y(u) = Y_0(1-u)^4 + Y_1 4u(1-u)^3 + Y_2 6u^2(1-u)^2 + Y_3 4u^3(1-u)^1 + Y_4 u^4 \dots \dots (iii)$$

Put those values in X_0, X_1, X_2, X_3, X_4 Values in equation 2:

$$\therefore X(u) = 1(1-u)^4 + 2 \times 4u(1-u)^3 + 4 \times 6u^2(1-u)^2 + 6 \times 4u^3(1-u)^1 + 8 \times u^4$$

$$\therefore X(u) = (1-u)^4 + 8u(1-u)^3 + 24u^2(1-u)^2 + 24u^3(1-u)^1 + 8u^4 \dots \dots (iv)$$

Put those values in Y_0, Y_1, Y_2, Y_3, Y_4 Values in equation 3:

$$\therefore Y(u) = 1(1-u)^4 + 3 \times 4u(1-u)^3 + 3 \times 6u^2(1-u)^2 + 4 \times 4u^3(1-u)^1 + 6u^4$$

$$\therefore Y(u) = (1-u)^4 + 12u(1-u)^3 + 18u^2(1-u)^2 + 16u^3(1-u)^1 + 6u^4 \dots \dots (v)$$

So the point P(u) on the cubic Bézier curve:

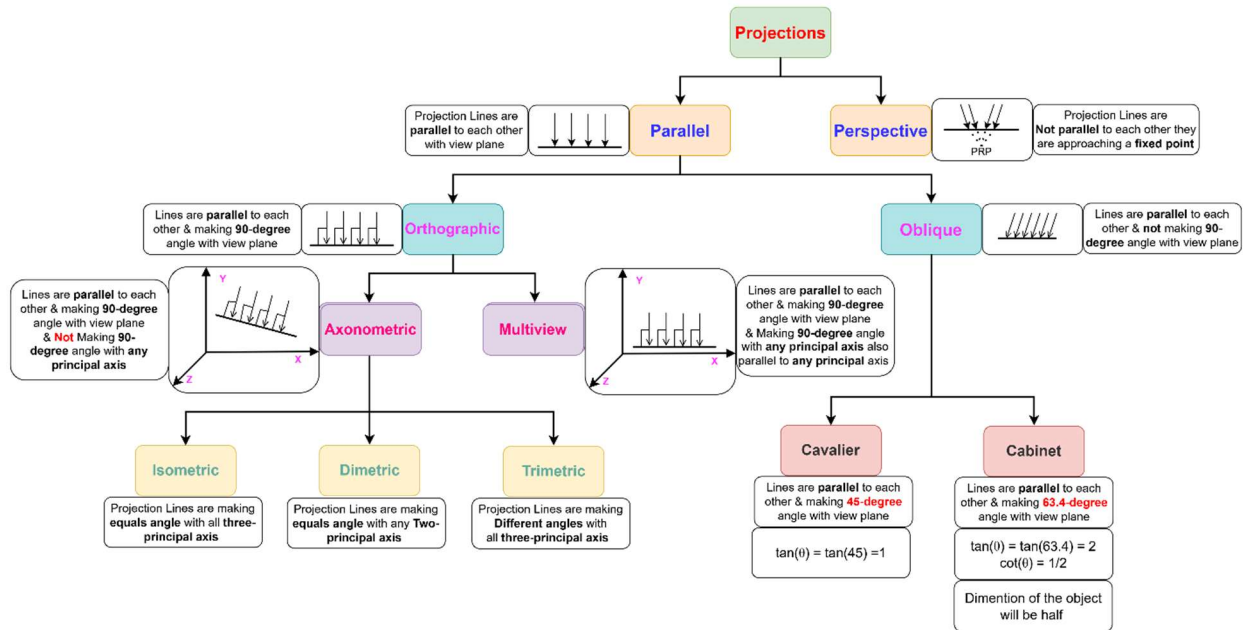
u	X	Y
0.0	1	1
0.2	2.0096	2.2112
0.4	3.3296	2.9712
0.6	4.8256	3.6832
0.8	6.4016	4.6352
1.0	8	6

PROJECTION

Projection

⇒ A **projection** is a method of mapping points from one space onto another, typically from a 3D space to a 2D plane. This process helps represent three-dimensional objects on a two-dimensional surface, such as a computer screen or a sheet of paper.

Projection Types



Project Link

Project Link: <https://github.com/MIHMahmudEli/GlutProject>

⚠ Important Note:

This project is **made purely for learning and educational purposes**. If you're a student, please **don't submit it exactly as it is** to your teacher or faculty. Instead, use it to learn the logic, understand how it works, and then build your own version with your own ideas.